Peizheng Li

Seattle, WA · Open to Relocation

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Summary

Ph.D. researcher in computer vision and machine learning with **4+ years of autonomous driving research at Mercedes-Benz R&D**, working on **3D world understanding with vision-language models**: lifting 2D vision-language semantics into open-world 3D occupancy, **scaling label-free 3D auto-labeling to production-fleet camera + LiDAR data** and shipping the distilled model to the vehicle team, spatially grounded VLMs for end-to-end (E2E) driving, and world-action models trained with counterfactual supervision. First-/co-first-author publications at **CVPR, ICCV, and IJCAI**; co-inventor on **3 Mercedes-Benz patents**.

Work Experience

Mercedes-Benz Group AG, Stuttgart, Germany | Scene Understanding Group, RD

Industrial Ph.D. Researcher — *Feb. 2023 - Present*

- Lead a research line on open-world 3D occupancy, label-free 3D pseudo-labeling, and spatial VLMs for driving; co-author SeFlow (scene flow), TQD-Track (3D tracking), and G2DP (planning); mentor 2 research interns and 1 master's student.
- **Open-vocabulary 3D occupancy auto-labeling** (project page): built and ran a label-free pipeline on production-fleet recordings (5 surround cameras + LiDAR) that fuses multi-frame LiDAR and 2D open-vocabulary semantics into 3D semantic voxel labels, then distills them into a single-frame model delivered to the vehicle team; labeled **1,500 episodes (30,000 keyframes) with no human annotation** within a $5K labeling budget (full 42,538-clip archive est. $199K); validated free space without ground truth by split-episode LiDAR replay (**0.11% false-free rate** vs. 2.18% for forward ray casting); traced ghost voxels to an int16 cache overflow (87% fewer after the fix); cut per-episode cost **36%**.
- **AGO** (ICCV 2025, 1st author, project page): open-world 3D occupancy trained **without 3D annotation**: 2D open-vocabulary masks lifted to 3D pseudo-labels via LiDAR projection, multi-frame aggregation, and ray casting; a **modality adapter aligns 3D embeddings to VLM image embeddings** for zero-shot and few-shot novel-class prediction; state of the art at publication on self-supervised Occ3D-nuScenes (**+4.09 mIoU**).
- **SpaceDrive** (CVPR 2026, co-first author, project page): E2E spatial VLA that replaces textual digit tokens with a unified 3D positional encoding, without dense BEV features; **state-of-the-art nuScenes open-loop planning** among VLA-based methods (0.32 m avg. L2 with ego status, 1.80 m without) and 78.02 driving score on Bench2Drive closed loop.
- **COURSE** (1st author, in preparation): world-action model for E2E driving trained on a **counterfactual causal tree**, where an action-conditioned latent transition model rolls logged scenes forward under alternative ego maneuvers to supervise the planner (NAVSIM).
- Scale training and evaluation on on-premise SLURM clusters and cloud (Flyte, Kubernetes, Azure, GCP); e.g., cut distillation step time **3.29**× by caching frozen vision-language teacher targets.

Master Student — *Apr. 2022 - Dec. 2022*

- **PowerBEV** (IJCAI 2023, co-first author, project page): camera-only E2E BEV perception and future instance prediction with Lift-Splat-Shoot view transformation and centripetal backward flow, replacing two-stage post-processing with one-step association; **62.5% / 39.3% IoU** (short / long range) on nuScenes.
- Designed a multi-stage temporal GCN for the Waymo Occupancy and Flow Prediction Challenge; achieved **74.2% Flow-Grounded Occupancy AUC** and ranked **4th** on the leaderboard.

Research Intern — *May 2021 - Oct. 2021*

- Studied contextual bias in 2D object detection and domain adaptation; showed disruptive context reduced mAP by **5.37%**, and built context-separation components for internal adaptation research.

Publications

11 publications (9 peer-reviewed); **3 first-/co-first-author papers at CVPR, ICCV, IJCAI**; **238 citations, h-index 7** (Google Scholar, 2026-10). **Bold underline** = author; * = equal contribution. Titles link to the official publication or arXiv.

SpaceDrive: Infusing Spatial Awareness into VLM-based Autonomous Driving *CVPR 2026*

Peizheng Li*, Zhenghao Zhang*, David Holtz, Hang Yu, Yutong Yang, Yuzhi Lai, Rui Song, Andreas Geiger, Andreas Zell

AGO: Adaptive Grounding for Open World 3D Occupancy Prediction *ICCV 2025*

Peizheng Li, Shuxiao Ding, You Zhou, Qingwen Zhang, Onat Inak, Larissa Triess, Niklas Hanselmann, Marius Cordts, Andreas Zell

SeFlow: A Self-Supervised Scene Flow Method in Autonomous Driving *ECCV 2024*
Qingwen Zhang, Yi Yang, **Peizheng Li**, Olov Andersson, Patric Jensfelt *(SOTA on self-supervised Argoverse 2 scene flow at publication)*

PowerBEV: A Powerful Yet Lightweight Framework for Instance Prediction in Bird's-Eye View *IJCAI 2023*
Peizheng Li*, Shuxiao Ding*, Xieyuanli Chen, Niklas Hanselmann, Marius Cordts, Juergen Gall

G2DP: Diffusion Planning with Spatio-Temporal Grid Guidance *IROS 2026*
Hang Yu, Ye Jin, Alessandro Canevaro, Julian Schmidt, Julian Jordan, **Peizheng Li**, Marc Kaufeld, Silvan Lindner, Johannes Betz, Wilhelm Stork

Shift & Drift: A Zero-Shot Benchmark for Generalizable and Robust Autonomous Driving Motion Planning *IROS 2026*
Alessandro Canevaro, Hang Yu, Julian Schmidt, **Peizheng Li**, Silvan Lindner, Wilhelm Stork, Georg Martius, Julian Jordan

TQD-Track: Temporal Query Denoising for 3D Multi-Object Tracking *BMVC 2026*
Yutong Yang*, Shuxiao Ding*, Mohammed Amine Bencheikh Lehocine, Julian Wiederer, Markus Braun, **Peizheng Li**, Juergen Gall, Bin Yang

FAM-HRI: Foundation-Model Assisted Multi-Modal Human-Robot Interaction Combining Gaze and Speech *T-ASE 2026*
Yuzhi Lai, Shenghai Yuan, **Peizheng Li**, Boya Zhang, Benjamin Kiefer, Tianchen Deng, Andreas Zell

SEER-VAR: Semantic Egocentric Environment Reasoner for Vehicle Augmented Reality *WACV 2027*
Yuzhi Lai, Shenghai Yuan, **Peizheng Li**, Jun Lou, Andreas Zell

Glance-Say: Multimodal Human-Robot Collaboration and Intent Recognition via Sticky Glance *arXiv 2026*
Yuzhi Lai, Shenghai Yuan, **Peizheng Li**, Benjamin Kiefer, Andreas Zell

Beyond Object Selection: Markerless Gaze-based Robot Placement at Arbitrary Positions *arXiv 2026*
Yuzhi Lai, William Marx, Shenghai Yuan, **Peizheng Li**, Zhuoyu Ran, Andreas Zell

Patents

A Powerful Yet Lightweight Framework for Instance Prediction in Bird's-Eye View *DE 102023001521 A1*
Shuxiao Ding, **Peizheng Li**, Niklas Hanselmann. Mercedes-Benz Group AG; published 2023, under examination.

An Open-World 3D Semantic Occupancy Prediction System Based on Grounded Training *DE 102025126853.3*
Peizheng Li, Shuxiao Ding. Mercedes-Benz Group AG; filed 2025, within confidentiality period.

A Spatially-Enhanced System Employing Explicit Positional Embedding for Vision-Language-Action-Based Autonomous Driving *DE 102026133405.9*
Zhenghao Zhang, **Peizheng Li**. Mercedes-Benz Group AG; filed 2026, examination requested.

Education

University of Tübingen *Feb. 2023 – Present*
Ph.D. Candidate in Computer Science
- Advisors: **Prof. Andreas Geiger** and **Prof. Andreas Zell**

University of Stuttgart *Apr. 2020 – Dec. 2022*
M.Sc. in Electromobility
- Grade: **1.6** (German scale, **1.0** best; approx. **3.7/4.0** US equivalent)
- Thesis: *End-to-End Agent Perception and Occupancy-Based Motion Prediction in Bird's-Eye View*

Tongji University *Sept. 2014 – Jul. 2019*
B.Eng. in Vehicle Engineering
- Grade: **4.5 / 5.0**
- Thesis: *Vehicle Detection and Tracking Based on Sensor Fusion*

Technical Skills

Programming: Python, C++, CUDA, C#, MATLAB, Bash
ML / Systems: PyTorch, TensorFlow, OpenMMLab, Distributed Training, Docker, Kubernetes, Flyte, W&B, Azure, GCP
Research Areas: 3D world understanding with VLMs (open-world 3D occupancy, 2D-to-3D semantic lifting), label-free 3D auto-labeling at fleet scale (multi-frame LiDAR aggregation, ray casting, visibility reasoning), BEV / scene flow / 3D tracking, VLM/VLA and world-action models for end-to-end driving
Languages: Chinese (native), English (professional), German (professional)

Service

Reviewer: CVPR 2025/2026, ICCV 2025, ECCV 2026, WACV 2027, ICRA 2026/2027, AAAI 2026/2027, IROS 2025/2026, IEEE T-ITS 2025
Organizer: 4th & 6th DriveX Workshops (CVPR 2026, ECCV 2026)